SG No. 227 O_{h}^{7} $Fd\bar{3}m$ [cubic]

* plus set: $+[0,\,0,\,0]$, $+\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$
* Wyckoff site: `8a`, site symmetry: `-43m`

Table 1: Wyckoff bond: `24a@8a`, bond symmetry: `-42.m`

No.	vector	center	mapping
1	$[X,\,0,\,0]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[1,-2,-3,4,-41,42,43,-44]`
2	$[0,\,X,\,0]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[5,-6,-7,8,-37,38,39,-40]`
3	$[0,\,0,\,X]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[9,-10,-11,12,45,-46,-47,48]`
4	$[0,\,X,\,0]$	$\left[\frac{7}{8},\,\frac{3}{8},\,\frac{3}{8}\right]$	`[13,-14,-15,16,-29,30,31,-32]`
5	$[X,\,0,\,0]$	$\left[\frac{7}{8},\,\frac{3}{8},\,\frac{3}{8}\right]$	`[17,-18,-19,20,-25,26,27,-28]`
6	$[0,\,0,\,-X]$	$\left[\frac{7}{8},\,\frac{3}{8},\,\frac{3}{8}\right]$	`[21,-22,-23,24,33,-34,-35,36]`

Table 2: Wyckoff bond: `32b@8a`, bond symmetry: `.3m`

No.	vector	center	mapping
1	$[X,\,X,\,X]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[1,5,9,38,43,48]`
2	$[-X,\,-X,\,X]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[2,7,12,37,41,45]`
3	$[-X,\,X,\,-X]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[3,8,10,39,44,46]`
4	$[X,\,-X,\,-X]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[4,6,11,40,42,47]`
5	$[X,\,X,\,-X]$	$\left[\frac{7}{8},\,\frac{3}{8},\,\frac{3}{8}\right]$	`[13,17,21,26,31,36]`
6	$[-X,\,-X,\,-X]$	$\left[\frac{7}{8},\,\frac{3}{8},\,\frac{3}{8}\right]$	`[14,19,24,25,29,33]`
7	$[X,\,-X,\,X]$	$\left[\frac{7}{8},\,\frac{3}{8},\,\frac{3}{8}\right]$	`[15,20,22,27,32,34]`
8	$[-X,\,X,\,X]$	$\left[\frac{7}{8},\,\frac{3}{8},\,\frac{3}{8}\right]$	`[16,18,23,28,30,35]`

Table 3: Wyckoff bond: `48c@8a`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X,\,X,\,0]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[1,-2,-37,38]`
2	$[-X,\,X,\,0]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[3,-4,39,-40]`
3	$[0,\,X,\,X]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[5,-6,-47,48]`
4	$[0,\,-X,\,X]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[7,-8,45,-46]`
5	$[X,\,0,\,X]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[9,-10,43,-44]`
6	$[X,\,0,\,-X]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[11,-12,-41,42]`
7	$[X,\,X,\,0]$	$\left[\frac{7}{8},\,\frac{3}{8},\,\frac{3}{8}\right]$	`[13,-14,-25,26]`
8	$[X,\,-X,\,0]$	$\left[\frac{7}{8},\,\frac{3}{8},\,\frac{3}{8}\right]$	`[15,-16,27,-28]`
9	$[X,\,0,\,-X]$	$\left[\frac{7}{8},\,\frac{3}{8},\,\frac{3}{8}\right]$	`[17,-18,-35,36]`
10	$[-X,\,0,\,-X]$	$\left[\frac{7}{8},\,\frac{3}{8},\,\frac{3}{8}\right]$	`[19,-20,33,-34]`
11	$[0,\,X,\,-X]$	$\left[\frac{7}{8},\,\frac{3}{8},\,\frac{3}{8}\right]$	`[21,-22,31,-32]`
12	$[0,\,X,\,X]$	$\left[\frac{7}{8},\,\frac{3}{8},\,\frac{3}{8}\right]$	`[23,-24,-29,30]`

Table 4: Wyckoff bond: `96d@8a`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X,\,X,\,Y]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[1,38]`

continued ...

Table 4

No.	vector	center	mapping
2	$[-X, -X, Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[2,37]
3	$[-X, X, -Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[3,39]
4	$[X, -X, -Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[4,40]
5	$[Y, X, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[5,48]
6	$[Y, -X, -X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[6,47]
7	$[-Y, -X, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[7,45]
8	$[-Y, X, -X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[8,46]
9	$[X, Y, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[9,43]
10	$[-X, Y, -X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[10,44]
11	$[X, -Y, -X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[11,42]
12	$[-X, -Y, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[12,41]
13	$[X, X, -Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[13,26]
14	$[-X, -X, -Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[14,25]
15	$[X, -X, Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[15,27]
16	$[-X, X, Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[16,28]
17	$[X, Y, -X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[17,36]
18	$[-X, Y, X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[18,35]
19	$[-X, -Y, -X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[19,33]
20	$[X, -Y, X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[20,34]
21	$[Y, X, -X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[21,31]
22	$[Y, -X, X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[22,32]
23	$[-Y, X, X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[23,30]
24	$[-Y, -X, -X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[24,29]

Table 5: Wyckoff bond: 96e@8a, bond symmetry: 2..

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[1,-2]
2	$[-X, Y, 0]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[3,-4]
3	$[0, X, Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[5,-6]
4	$[0, -X, Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[7,-8]
5	$[Y, 0, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[9,-10]
6	$[Y, 0, -X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[11,-12]
7	$[Y, X, 0]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[13,-14]
8	$[Y, -X, 0]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[15,-16]
9	$[X, 0, -Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[17,-18]
10	$[-X, 0, -Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[19,-20]
11	$[0, Y, -X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[21,-22]
12	$[0, Y, X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[23,-24]
13	$[-X, -Y, 0]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[25,-26]
14	$[X, -Y, 0]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[27,-28]
15	$[0, -X, -Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[29,-30]
16	$[0, X, -Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[31,-32]
17	$[-Y, 0, -X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[33,-34]
18	$[-Y, 0, X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[35,-36]

continued ...

Table 5

No.	vector	center	mapping
19	$[-Y, -X, 0]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[37,-38]`
20	$[-Y, X, 0]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[39,-40]`
21	$[-X, 0, Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[41,-42]`
22	$[X, 0, Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[43,-44]`
23	$[0, -Y, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[45,-46]`
24	$[0, -Y, -X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[47,-48]`

Table 6: Wyckoff bond: `192f@8a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[2]`
3	$[-X, Y, -Z]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[3]`
4	$[X, -Y, -Z]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[4]`
5	$[Z, X, Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[5]`
6	$[Z, -X, -Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[6]`
7	$[-Z, -X, Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[7]`
8	$[-Z, X, -Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[8]`
9	$[Y, Z, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[9]`
10	$[-Y, Z, -X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[10]`
11	$[Y, -Z, -X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[11]`
12	$[-Y, -Z, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[12]`
13	$[Y, X, -Z]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[13]`
14	$[-Y, -X, -Z]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[14]`
15	$[Y, -X, Z]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[15]`
16	$[-Y, X, Z]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[16]`
17	$[X, Z, -Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[17]`
18	$[-X, Z, Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[18]`
19	$[-X, -Z, -Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[19]`
20	$[X, -Z, Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[20]`
21	$[Z, Y, -X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[21]`
22	$[Z, -Y, X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[22]`
23	$[-Z, Y, X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[23]`
24	$[-Z, -Y, -X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[24]`
25	$[-X, -Y, -Z]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[25]`
26	$[X, Y, -Z]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[26]`
27	$[X, -Y, Z]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[27]`
28	$[-X, Y, Z]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[28]`
29	$[-Z, -X, -Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[29]`
30	$[-Z, X, Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[30]`
31	$[Z, X, -Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[31]`
32	$[Z, -X, Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[32]`
33	$[-Y, -Z, -X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[33]`
34	$[Y, -Z, X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[34]`
35	$[-Y, Z, X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	`[35]`

continued ...

Table 6

No.	vector	center	mapping
36	$[Y, Z, -X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}]$	[36]
37	$[-Y, -X, Z]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[37]
38	$[Y, X, Z]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[38]
39	$[-Y, X, -Z]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[39]
40	$[Y, -X, -Z]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[40]
41	$[-X, -Z, Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[41]
42	$[X, -Z, -Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[42]
43	$[X, Z, Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[43]
44	$[-X, Z, -Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[44]
45	$[-Z, -Y, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[45]
46	$[-Z, Y, -X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[46]
47	$[Z, -Y, -X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[47]
48	$[Z, Y, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	[48]

* Wyckoff site: 8b, site symmetry: -43m

Table 7: Wyckoff bond: 24a@8b, bond symmetry: -42.m

No.	vector	center	mapping
1	$[X, 0, 0]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	[1,-2,-3,4,-41,42,43,-44]
2	$[0, X, 0]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	[5,-6,-7,8,-37,38,39,-40]
3	$[0, 0, X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	[9,-10,-11,12,45,-46,-47,48]
4	$[0, X, 0]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	[13,-14,-15,16,-29,30,31,-32]
5	$[X, 0, 0]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	[17,-18,-19,20,-25,26,27,-28]
6	$[0, 0, -X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	[21,-22,-23,24,33,-34,-35,36]

Table 8: Wyckoff bond: 32b@8b, bond symmetry: .3m

No.	vector	center	mapping
1	$[X, X, X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	[1,5,9,38,43,48]
2	$[-X, -X, X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	[2,7,12,37,41,45]
3	$[-X, X, -X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	[3,8,10,39,44,46]
4	$[X, -X, -X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	[4,6,11,40,42,47]
5	$[X, X, -X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	[13,17,21,26,31,36]
6	$[-X, -X, -X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	[14,19,24,25,29,33]
7	$[X, -X, X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	[15,20,22,27,32,34]
8	$[-X, X, X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	[16,18,23,28,30,35]

Table 9: Wyckoff bond: 48c@8b, bond symmetry: 2.mm

No.	vector	center	mapping
1	$[X, X, 0]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	[1,-2,-37,38]
2	$[-X, X, 0]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	[3,-4,39,-40]
3	$[0, X, X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	[5,-6,-47,48]

continued ...

Table 9

No.	vector	center	mapping
4	$[0, -X, X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[7,-8,45,-46]`
5	$[X, 0, X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[9,-10,43,-44]`
6	$[X, 0, -X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[11,-12,-41,42]`
7	$[X, X, 0]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[13,-14,-25,26]`
8	$[X, -X, 0]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[15,-16,27,-28]`
9	$[X, 0, -X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[17,-18,-35,36]`
10	$[-X, 0, -X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[19,-20,33,-34]`
11	$[0, X, -X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[21,-22,31,-32]`
12	$[0, X, X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[23,-24,-29,30]`

Table 10: Wyckoff bond: `96d@8b`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Y]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[1,38]`
2	$[-X, -X, Y]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[2,37]`
3	$[-X, X, -Y]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[3,39]`
4	$[X, -X, -Y]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[4,40]`
5	$[Y, X, X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[5,48]`
6	$[Y, -X, -X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[6,47]`
7	$[-Y, -X, X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[7,45]`
8	$[-Y, X, -X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[8,46]`
9	$[X, Y, X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[9,43]`
10	$[-X, Y, -X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[10,44]`
11	$[X, -Y, -X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[11,42]`
12	$[-X, -Y, X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[12,41]`
13	$[X, X, -Y]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[13,26]`
14	$[-X, -X, -Y]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[14,25]`
15	$[X, -X, Y]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[15,27]`
16	$[-X, X, Y]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[16,28]`
17	$[X, Y, -X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[17,36]`
18	$[-X, Y, X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[18,35]`
19	$[-X, -Y, -X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[19,33]`
20	$[X, -Y, X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[20,34]`
21	$[Y, X, -X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[21,31]`
22	$[Y, -X, X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[22,32]`
23	$[-Y, X, X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[23,30]`
24	$[-Y, -X, -X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[24,29]`

Table 11: Wyckoff bond: `96e@8b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[1,-2]`
2	$[-X, Y, 0]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[3,-4]`
3	$[0, X, Y]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[5,-6]`

continued ...

Table 11

No.	vector	center	mapping
4	$[0, -X, Y]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[7,-8]`
5	$[Y, 0, X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[9,-10]`
6	$[Y, 0, -X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[11,-12]`
7	$[Y, X, 0]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[13,-14]`
8	$[Y, -X, 0]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[15,-16]`
9	$[X, 0, -Y]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[17,-18]`
10	$[-X, 0, -Y]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[19,-20]`
11	$[0, Y, -X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[21,-22]`
12	$[0, Y, X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[23,-24]`
13	$[-X, -Y, 0]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[25,-26]`
14	$[X, -Y, 0]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[27,-28]`
15	$[0, -X, -Y]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[29,-30]`
16	$[0, X, -Y]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[31,-32]`
17	$[-Y, 0, -X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[33,-34]`
18	$[-Y, 0, X]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[35,-36]`
19	$[-Y, -X, 0]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[37,-38]`
20	$[-Y, X, 0]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[39,-40]`
21	$[-X, 0, Y]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[41,-42]`
22	$[X, 0, Y]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[43,-44]`
23	$[0, -Y, X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[45,-46]`
24	$[0, -Y, -X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[47,-48]`

Table 12: Wyckoff bond: `192f@8b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[2]`
3	$[-X, Y, -Z]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[3]`
4	$[X, -Y, -Z]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[4]`
5	$[Z, X, Y]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[5]`
6	$[Z, -X, -Y]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[6]`
7	$[-Z, -X, Y]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[7]`
8	$[-Z, X, -Y]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[8]`
9	$[Y, Z, X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[9]`
10	$[-Y, Z, -X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[10]`
11	$[Y, -Z, -X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[11]`
12	$[-Y, -Z, X]$	$[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}]$	`[12]`
13	$[Y, X, -Z]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[13]`
14	$[-Y, -X, -Z]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[14]`
15	$[Y, -X, Z]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[15]`
16	$[-Y, X, Z]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[16]`
17	$[X, Z, -Y]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[17]`
18	$[-X, Z, Y]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[18]`
19	$[-X, -Z, -Y]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[19]`
20	$[X, -Z, Y]$	$[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}]$	`[20]`

continued ...

Table 12

No.	vector	center	mapping
21	$[Z, Y, -X]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[25]
26	$[X, Y, -Z]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[26]
27	$[X, -Y, Z]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[27]
28	$[-X, Y, Z]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[30]
31	$[Z, X, -Y]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[31]
32	$[Z, -X, Y]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[33]
34	$[Y, -Z, X]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[34]
35	$[-Y, Z, X]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[35]
36	$[Y, Z, -X]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[42]
43	$[X, Z, Y]$	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[47]
48	$[Z, Y, X]$	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[48]

* Wyckoff site: 16c, site symmetry: .-3m

Table 13: Wyckoff bond: 16a@16c, bond symmetry: .-3m

No.	vector	center	mapping
1	$[X, X, X]$	[0, 0, 0]	[1,5,9,-14,-19,-24,-25,-29,-33,38,43,48]
2	$[-X, -X, X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[2,7,12,-13,-17,-21,-26,-31,-36,37,41,45]
3	$[-X, X, -X]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[3,8,10,-15,-20,-22,-27,-32,-34,39,44,46]
4	$[X, -X, -X]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[4,6,11,-16,-18,-23,-28,-30,-35,40,42,47]

Table 14: Wyckoff bond: 48b@16c, bond symmetry: ..2/m

No.	vector	center	mapping
1	$[X, X, Y]$	[0, 0, 0]	[1,-14,-25,38]
2	$[-X, -X, Y]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[2,-13,-26,37]
3	$[-X, X, -Y]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[3,-15,-27,39]
4	$[X, -X, -Y]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[4,-16,-28,40]

continued ...

Table 14

No.	vector	center	mapping
5	$[Y, X, X]$	$[0, 0, 0]$	`[5,-24,-29,48]`
6	$[Y, -X, -X]$	$[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}]$	`[6,-23,-30,47]`
7	$[-Y, -X, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}]$	`[7,-21,-31,45]`
8	$[-Y, X, -X]$	$[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}]$	`[8,-22,-32,46]`
9	$[X, Y, X]$	$[0, 0, 0]$	`[9,-19,-33,43]`
10	$[-X, Y, -X]$	$[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}]$	`[10,-20,-34,44]`
11	$[X, -Y, -X]$	$[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}]$	`[11,-18,-35,42]`
12	$[-X, -Y, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}]$	`[12,-17,-36,41]`

Table 15: Wyckoff bond: `48c@16c`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[0, 0, 0]$	`[1,14,-25,-38]`
2	$[-X, X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}]$	`[2,13,-26,-37]`
3	$[-X, -X, 0]$	$[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}]$	`[3,15,-27,-39]`
4	$[X, X, 0]$	$[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}]$	`[4,16,-28,-40]`
5	$[0, X, -X]$	$[0, 0, 0]$	`[5,24,-29,-48]`
6	$[0, -X, X]$	$[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}]$	`[6,23,-30,-47]`
7	$[0, -X, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}]$	`[7,21,-31,-45]`
8	$[0, X, X]$	$[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}]$	`[8,22,-32,-46]`
9	$[-X, 0, X]$	$[0, 0, 0]$	`[9,19,-33,-43]`
10	$[X, 0, -X]$	$[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}]$	`[10,20,-34,-44]`
11	$[-X, 0, -X]$	$[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}]$	`[11,18,-35,-42]`
12	$[X, 0, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}]$	`[12,17,-36,-41]`

Table 16: Wyckoff bond: `96d@16c`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-25]`
2	$[-X, -Y, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}]$	`[2,-26]`
3	$[-X, Y, -Z]$	$[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}]$	`[3,-27]`
4	$[X, -Y, -Z]$	$[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}]$	`[4,-28]`
5	$[Z, X, Y]$	$[0, 0, 0]$	`[5,-29]`
6	$[Z, -X, -Y]$	$[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}]$	`[6,-30]`
7	$[-Z, -X, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}]$	`[7,-31]`
8	$[-Z, X, -Y]$	$[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}]$	`[8,-32]`
9	$[Y, Z, X]$	$[0, 0, 0]$	`[9,-33]`
10	$[-Y, Z, -X]$	$[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}]$	`[10,-34]`
11	$[Y, -Z, -X]$	$[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}]$	`[11,-35]`
12	$[-Y, -Z, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}]$	`[12,-36]`
13	$[Y, X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}]$	`[13,-37]`
14	$[-Y, -X, -Z]$	$[0, 0, 0]$	`[14,-38]`
15	$[Y, -X, Z]$	$[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}]$	`[15,-39]`
16	$[-Y, X, Z]$	$[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}]$	`[16,-40]`

continued ...

Table 16

No.	vector	center	mapping
17	$[X, Z, -Y]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[17,-41]`
18	$[-X, Z, Y]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[18,-42]`
19	$[-X, -Z, -Y]$	$[0, 0, 0]$	`[19,-43]`
20	$[X, -Z, Y]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[20,-44]`
21	$[Z, Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[21,-45]`
22	$[Z, -Y, X]$	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[22,-46]`
23	$[-Z, Y, X]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[23,-47]`
24	$[-Z, -Y, -X]$	$[0, 0, 0]$	`[24,-48]`

* Wyckoff site: `16d`, site symmetry: `.-3m`

Table 17: Wyckoff bond: `16a@16d`, bond symmetry: `.-3m`

No.	vector	center	mapping
1	$[X, X, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,5,9,-14,-19,-24,-25,-29,-33,38,43,48]`
2	$[-X, -X, X]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[2,7,12,-13,-17,-21,-26,-31,-36,37,41,45]`
3	$[-X, X, -X]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[3,8,10,-15,-20,-22,-27,-32,-34,39,44,46]`
4	$[X, -X, -X]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[4,6,11,-16,-18,-23,-28,-30,-35,40,42,47]`

Table 18: Wyckoff bond: `48b@16d`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-14,-25,38]`
2	$[-X, -X, Y]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[2,-13,-26,37]`
3	$[-X, X, -Y]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[3,-15,-27,39]`
4	$[X, -X, -Y]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[4,-16,-28,40]`
5	$[Y, X, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-24,-29,48]`
6	$[Y, -X, -X]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[6,-23,-30,47]`
7	$[-Y, -X, X]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[7,-21,-31,45]`
8	$[-Y, X, -X]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[8,-22,-32,46]`
9	$[X, Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,-19,-33,43]`
10	$[-X, Y, -X]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[10,-20,-34,44]`
11	$[X, -Y, -X]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[11,-18,-35,42]`
12	$[-X, -Y, X]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[12,-17,-36,41]`

Table 19: Wyckoff bond: `48c@16d`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,14,-25,-38]`
2	$[-X, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[2,13,-26,-37]`
3	$[-X, -X, 0]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[3,15,-27,-39]`
4	$[X, X, 0]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[4,16,-28,-40]`
5	$[0, X, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,24,-29,-48]`
6	$[0, -X, X]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[6,23,-30,-47]`

continued ...

Table 19

No.	vector	center	mapping
7	$[0, -X, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[7,21,-31,-45]`
8	$[0, X, X]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[8,22,-32,-46]`
9	$[-X, 0, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,19,-33,-43]`
10	$[X, 0, -X]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[10,20,-34,-44]`
11	$[-X, 0, -X]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[11,18,-35,-42]`
12	$[X, 0, X]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[12,17,-36,-41]`

Table 20: Wyckoff bond: `96d@16d`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-25]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[2,-26]`
3	$[-X, Y, -Z]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[3,-27]`
4	$[X, -Y, -Z]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[4,-28]`
5	$[Z, X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-29]`
6	$[Z, -X, -Y]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[6,-30]`
7	$[-Z, -X, Y]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[7,-31]`
8	$[-Z, X, -Y]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[8,-32]`
9	$[Y, Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,-33]`
10	$[-Y, Z, -X]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[10,-34]`
11	$[Y, -Z, -X]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[11,-35]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[12,-36]`
13	$[Y, X, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[13,-37]`
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[14,-38]`
15	$[Y, -X, Z]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[15,-39]`
16	$[-Y, X, Z]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[16,-40]`
17	$[X, Z, -Y]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[17,-41]`
18	$[-X, Z, Y]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[18,-42]`
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[19,-43]`
20	$[X, -Z, Y]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[20,-44]`
21	$[Z, Y, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[21,-45]`
22	$[Z, -Y, X]$	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[22,-46]`
23	$[-Z, Y, X]$	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[23,-47]`
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[24,-48]`

* Wyckoff site: `32e`, site symmetry: `.3m`

Table 21: Wyckoff bond: `32a@32e`, bond symmetry: `.3m`

No.	vector	center	mapping
1	$[X, X, X]$	$[x, x, x]$	`[1,5,9,38,43,48]`
2	$[-X, -X, X]$	$\left[\frac{3}{4} - x, \frac{1}{4} - x, x + \frac{1}{2}\right]$	`[2,7,12,37,41,45]`
3	$[-X, X, -X]$	$\left[\frac{1}{4} - x, x + \frac{1}{2}, \frac{3}{4} - x\right]$	`[3,8,10,39,44,46]`
4	$[X, -X, -X]$	$\left[x + \frac{1}{2}, \frac{3}{4} - x, \frac{1}{4} - x\right]$	`[4,6,11,40,42,47]`
5	$[X, X, -X]$	$\left[x + \frac{3}{4}, x + \frac{1}{4}, \frac{1}{2} - x\right]$	`[13,17,21,26,31,36]`
6	$[-X, -X, -X]$	$[-x, -x, -x]$	`[14,19,24,25,29,33]`

continued ...

Table 21

No.	vector	center	mapping
7	$[X, -X, X]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	`[15,20,22,27,32,34]`
8	$[-X, X, X]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	`[16,18,23,28,30,35]`

Table 22: Wyckoff bond: `96b@32e`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Y]$	$[x, x, x]$	`[1,38]`
2	$[-X, -X, Y]$	$\left[\frac{3}{4}-x, \frac{1}{4}-x, x+\frac{1}{2}\right]$	`[2,37]`
3	$[-X, X, -Y]$	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-x\right]$	`[3,39]`
4	$[X, -X, -Y]$	$\left[x+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-x\right]$	`[4,40]`
5	$[Y, X, X]$	$[x, x, x]$	`[5,48]`
6	$[Y, -X, -X]$	$\left[x+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-x\right]$	`[6,47]`
7	$[-Y, -X, X]$	$\left[\frac{3}{4}-x, \frac{1}{4}-x, x+\frac{1}{2}\right]$	`[7,45]`
8	$[-Y, X, -X]$	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-x\right]$	`[8,46]`
9	$[X, Y, X]$	$[x, x, x]$	`[9,43]`
10	$[-X, Y, -X]$	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-x\right]$	`[10,44]`
11	$[X, -Y, -X]$	$\left[x+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-x\right]$	`[11,42]`
12	$[-X, -Y, X]$	$\left[\frac{3}{4}-x, \frac{1}{4}-x, x+\frac{1}{2}\right]$	`[12,41]`
13	$[X, X, -Y]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	`[13,26]`
14	$[-X, -X, -Y]$	$[-x, -x, -x]$	`[14,25]`
15	$[X, -X, Y]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	`[15,27]`
16	$[-X, X, Y]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	`[16,28]`
17	$[X, Y, -X]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	`[17,36]`
18	$[-X, Y, X]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	`[18,35]`
19	$[-X, -Y, -X]$	$[-x, -x, -x]$	`[19,33]`
20	$[X, -Y, X]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	`[20,34]`
21	$[Y, X, -X]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	`[21,31]`
22	$[Y, -X, X]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	`[22,32]`
23	$[-Y, X, X]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	`[23,30]`
24	$[-Y, -X, -X]$	$[-x, -x, -x]$	`[24,29]`

Table 23: Wyckoff bond: `96c@32e`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[x, x, x]$	`[1,-38]`
2	$[-X, X, 0]$	$\left[\frac{3}{4}-x, \frac{1}{4}-x, x+\frac{1}{2}\right]$	`[2,-37]`
3	$[-X, -X, 0]$	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-x\right]$	`[3,-39]`
4	$[X, X, 0]$	$\left[x+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-x\right]$	`[4,-40]`
5	$[0, X, -X]$	$[x, x, x]$	`[5,-48]`
6	$[0, -X, X]$	$\left[x+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-x\right]$	`[6,-47]`
7	$[0, -X, -X]$	$\left[\frac{3}{4}-x, \frac{1}{4}-x, x+\frac{1}{2}\right]$	`[7,-45]`
8	$[0, X, X]$	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-x\right]$	`[8,-46]`
9	$[-X, 0, X]$	$[x, x, x]$	`[9,-43]`
10	$[X, 0, -X]$	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-x\right]$	`[10,-44]`

continued ...

Table 23

No.	vector	center	mapping
11	$[-X, 0, -X]$	$\left[x+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-x\right]$	`[11,-42]`
12	$[X, 0, X]$	$\left[\frac{3}{4}-x, \frac{1}{4}-x, x+\frac{1}{2}\right]$	`[12,-41]`
13	$[-X, X, 0]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	`[13,-26]`
14	$[X, -X, 0]$	$[-x, -x, -x]$	`[14,-25]`
15	$[-X, -X, 0]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	`[15,-27]`
16	$[X, X, 0]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	`[16,-28]`
17	$[X, 0, X]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	`[17,-36]`
18	$[-X, 0, -X]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	`[18,-35]`
19	$[-X, 0, X]$	$[-x, -x, -x]$	`[19,-33]`
20	$[X, 0, -X]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	`[20,-34]`
21	$[0, -X, -X]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	`[21,-31]`
22	$[0, X, X]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	`[22,-32]`
23	$[0, -X, X]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	`[23,-30]`
24	$[0, X, -X]$	$[-x, -x, -x]$	`[24,-29]`

Table 24: Wyckoff bond: `192d@32e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, x]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{3}{4}-x, \frac{1}{4}-x, x+\frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-x\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[x+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-x\right]$	`[4]`
5	$[Z, X, Y]$	$[x, x, x]$	`[5]`
6	$[Z, -X, -Y]$	$\left[x+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-x\right]$	`[6]`
7	$[-Z, -X, Y]$	$\left[\frac{3}{4}-x, \frac{1}{4}-x, x+\frac{1}{2}\right]$	`[7]`
8	$[-Z, X, -Y]$	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-x\right]$	`[8]`
9	$[Y, Z, X]$	$[x, x, x]$	`[9]`
10	$[-Y, Z, -X]$	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-x\right]$	`[10]`
11	$[Y, -Z, -X]$	$\left[x+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-x\right]$	`[11]`
12	$[-Y, -Z, X]$	$\left[\frac{3}{4}-x, \frac{1}{4}-x, x+\frac{1}{2}\right]$	`[12]`
13	$[Y, X, -Z]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	`[13]`
14	$[-Y, -X, -Z]$	$[-x, -x, -x]$	`[14]`
15	$[Y, -X, Z]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	`[15]`
16	$[-Y, X, Z]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	`[16]`
17	$[X, Z, -Y]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	`[17]`
18	$[-X, Z, Y]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	`[18]`
19	$[-X, -Z, -Y]$	$[-x, -x, -x]$	`[19]`
20	$[X, -Z, Y]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	`[20]`
21	$[Z, Y, -X]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	`[21]`
22	$[Z, -Y, X]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	`[22]`
23	$[-Z, Y, X]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	`[23]`
24	$[-Z, -Y, -X]$	$[-x, -x, -x]$	`[24]`
25	$[-X, -Y, -Z]$	$[-x, -x, -x]$	`[25]`
26	$[X, Y, -Z]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	`[26]`
27	$[X, -Y, Z]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	`[27]`

continued ...

Table 24

No.	vector	center	mapping
28	$[-X, Y, Z]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	[28]
29	$[-Z, -X, -Y]$	$[-x, -x, -x]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	[30]
31	$[Z, X, -Y]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	[31]
32	$[Z, -X, Y]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	[32]
33	$[-Y, -Z, -X]$	$[-x, -x, -x]$	[33]
34	$[Y, -Z, X]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	[34]
35	$[-Y, Z, X]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	[35]
36	$[Y, Z, -X]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{3}{4}-x, \frac{1}{4}-x, x+\frac{1}{2}\right]$	[37]
38	$[Y, X, Z]$	$[x, x, x]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-x\right]$	[39]
40	$[Y, -X, -Z]$	$\left[x+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-x\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{3}{4}-x, \frac{1}{4}-x, x+\frac{1}{2}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-x\right]$	[42]
43	$[X, Z, Y]$	$[x, x, x]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-x\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{3}{4}-x, \frac{1}{4}-x, x+\frac{1}{2}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-x\right]$	[46]
47	$[Z, -Y, -X]$	$\left[x+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-x\right]$	[47]
48	$[Z, Y, X]$	$[x, x, x]$	[48]

* Wyckoff site: 48f, site symmetry: 2.mm

Table 25: Wyckoff bond: 48a@48f, bond symmetry: 2.mm

No.	vector	center	mapping
1	$[0, X, X]$	$\left[x, \frac{1}{8}, \frac{1}{8}\right]$	[1,-4,-42,43]
2	$[0, -X, X]$	$\left[\frac{3}{4}-x, \frac{1}{8}, \frac{5}{8}\right]$	[2,-3,41,-44]
3	$[X, 0, X]$	$\left[\frac{1}{8}, x, \frac{1}{8}\right]$	[5,-8,38,-39]
4	$[X, 0, -X]$	$\left[\frac{5}{8}, \frac{3}{4}-x, \frac{1}{8}\right]$	[6,-7,-37,40]
5	$[X, X, 0]$	$\left[\frac{1}{8}, \frac{1}{8}, x\right]$	[9,-12,-45,48]
6	$[-X, X, 0]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{3}{4}-x\right]$	[10,-11,46,-47]
7	$[X, 0, -X]$	$\left[\frac{7}{8}, x+\frac{1}{4}, \frac{3}{8}\right]$	[13,-16,-30,31]
8	$[-X, 0, -X]$	$\left[\frac{7}{8}, -x, \frac{7}{8}\right]$	[14,-15,29,-32]
9	$[0, X, -X]$	$\left[x+\frac{3}{4}, \frac{3}{8}, \frac{3}{8}\right]$	[17,-20,26,-27]
10	$[0, X, X]$	$\left[\frac{1}{2}-x, \frac{7}{8}, \frac{3}{8}\right]$	[18,-19,-25,28]
11	$[X, X, 0]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{1}{2}-x\right]$	[21,-24,-33,36]
12	$[X, -X, 0]$	$\left[\frac{3}{8}, \frac{3}{8}, x+\frac{3}{4}\right]$	[22,-23,34,-35]

Table 26: Wyckoff bond: 48b@48f, bond symmetry: 2.mm

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{1}{8}, \frac{1}{8}\right]$	[1,4,42,43]
2	$[-X, 0, 0]$	$\left[\frac{3}{4}-x, \frac{1}{8}, \frac{5}{8}\right]$	[2,3,41,44]
3	$[0, X, 0]$	$\left[\frac{1}{8}, x, \frac{1}{8}\right]$	[5,8,38,39]

continued ...

Table 26

No.	vector	center	mapping
4	$[0, -X, 0]$	$\left[\frac{5}{8}, \frac{3}{4} - x, \frac{1}{8}\right]$	`[6,7,37,40]`
5	$[0, 0, X]$	$\left[\frac{1}{8}, \frac{1}{8}, x\right]$	`[9,12,45,48]`
6	$[0, 0, -X]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{3}{4} - x\right]$	`[10,11,46,47]`
7	$[0, X, 0]$	$\left[\frac{7}{8}, x + \frac{1}{4}, \frac{3}{8}\right]$	`[13,16,30,31]`
8	$[0, -X, 0]$	$\left[\frac{7}{8}, -x, \frac{7}{8}\right]$	`[14,15,29,32]`
9	$[X, 0, 0]$	$\left[x + \frac{3}{4}, \frac{3}{8}, \frac{3}{8}\right]$	`[17,20,26,27]`
10	$[-X, 0, 0]$	$\left[\frac{1}{2} - x, \frac{7}{8}, \frac{3}{8}\right]$	`[18,19,25,28]`
11	$[0, 0, -X]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{1}{2} - x\right]$	`[21,24,33,36]`
12	$[0, 0, X]$	$\left[\frac{3}{8}, \frac{3}{8}, x + \frac{3}{4}\right]$	`[22,23,34,35]`

Table 27: Wyckoff bond: `96c@48f`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[Y, X, X]$	$\left[x, \frac{1}{8}, \frac{1}{8}\right]$	`[1,43]`
2	$[-Y, -X, X]$	$\left[\frac{3}{4} - x, \frac{1}{8}, \frac{5}{8}\right]$	`[2,41]`
3	$[-Y, X, -X]$	$\left[\frac{3}{4} - x, \frac{1}{8}, \frac{5}{8}\right]$	`[3,44]`
4	$[Y, -X, -X]$	$\left[x, \frac{1}{8}, \frac{1}{8}\right]$	`[4,42]`
5	$[X, Y, X]$	$\left[\frac{1}{8}, x, \frac{1}{8}\right]$	`[5,38]`
6	$[X, -Y, -X]$	$\left[\frac{5}{8}, \frac{3}{4} - x, \frac{1}{8}\right]$	`[6,40]`
7	$[-X, -Y, X]$	$\left[\frac{5}{8}, \frac{3}{4} - x, \frac{1}{8}\right]$	`[7,37]`
8	$[-X, Y, -X]$	$\left[\frac{1}{8}, x, \frac{1}{8}\right]$	`[8,39]`
9	$[X, X, Y]$	$\left[\frac{1}{8}, \frac{1}{8}, x\right]$	`[9,48]`
10	$[-X, X, -Y]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{3}{4} - x\right]$	`[10,46]`
11	$[X, -X, -Y]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{3}{4} - x\right]$	`[11,47]`
12	$[-X, -X, Y]$	$\left[\frac{1}{8}, \frac{1}{8}, x\right]$	`[12,45]`
13	$[X, Y, -X]$	$\left[\frac{7}{8}, x + \frac{1}{4}, \frac{3}{8}\right]$	`[13,31]`
14	$[-X, -Y, -X]$	$\left[\frac{7}{8}, -x, \frac{7}{8}\right]$	`[14,29]`
15	$[X, -Y, X]$	$\left[\frac{7}{8}, -x, \frac{7}{8}\right]$	`[15,32]`
16	$[-X, Y, X]$	$\left[\frac{7}{8}, x + \frac{1}{4}, \frac{3}{8}\right]$	`[16,30]`
17	$[Y, X, -X]$	$\left[x + \frac{3}{4}, \frac{3}{8}, \frac{3}{8}\right]$	`[17,26]`
18	$[-Y, X, X]$	$\left[\frac{1}{2} - x, \frac{7}{8}, \frac{3}{8}\right]$	`[18,28]`
19	$[-Y, -X, -X]$	$\left[\frac{1}{2} - x, \frac{7}{8}, \frac{3}{8}\right]$	`[19,25]`
20	$[Y, -X, X]$	$\left[x + \frac{3}{4}, \frac{3}{8}, \frac{3}{8}\right]$	`[20,27]`
21	$[X, X, -Y]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{1}{2} - x\right]$	`[21,36]`
22	$[X, -X, Y]$	$\left[\frac{3}{8}, \frac{3}{8}, x + \frac{3}{4}\right]$	`[22,34]`
23	$[-X, X, Y]$	$\left[\frac{3}{8}, \frac{3}{8}, x + \frac{3}{4}\right]$	`[23,35]`
24	$[-X, -X, -Y]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{1}{2} - x\right]$	`[24,33]`

Table 28: Wyckoff bond: `96d@48f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[x, \frac{1}{8}, \frac{1}{8}\right]$	`[1,-4]`
2	$[0, -X, Y]$	$\left[\frac{3}{4} - x, \frac{1}{8}, \frac{5}{8}\right]$	`[2,-3]`
3	$[Y, 0, X]$	$\left[\frac{1}{8}, x, \frac{1}{8}\right]$	`[5,-8]`

continued ...

Table 28

No.	vector	center	mapping
4	$[Y, 0, -X]$	$\left[\frac{5}{8}, \frac{3}{4}-x, \frac{1}{8}\right]$	[6,-7]
5	$[X, Y, 0]$	$\left[\frac{1}{8}, \frac{1}{8}, x\right]$	[9,-12]
6	$[-X, Y, 0]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{3}{4}-x\right]$	[10,-11]
7	$[X, 0, -Y]$	$\left[\frac{7}{8}, x+\frac{1}{4}, \frac{3}{8}\right]$	[13,-16]
8	$[-X, 0, -Y]$	$\left[\frac{7}{8}, -x, \frac{7}{8}\right]$	[14,-15]
9	$[0, Y, -X]$	$\left[x+\frac{3}{4}, \frac{3}{8}, \frac{3}{8}\right]$	[17,-20]
10	$[0, Y, X]$	$\left[\frac{1}{2}-x, \frac{7}{8}, \frac{3}{8}\right]$	[18,-19]
11	$[Y, X, 0]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{1}{2}-x\right]$	[21,-24]
12	$[Y, -X, 0]$	$\left[\frac{3}{8}, \frac{3}{8}, x+\frac{3}{4}\right]$	[22,-23]
13	$[0, -X, -Y]$	$\left[\frac{1}{2}-x, \frac{7}{8}, \frac{3}{8}\right]$	[25,-28]
14	$[0, X, -Y]$	$\left[x+\frac{3}{4}, \frac{3}{8}, \frac{3}{8}\right]$	[26,-27]
15	$[-Y, 0, -X]$	$\left[\frac{7}{8}, -x, \frac{7}{8}\right]$	[29,-32]
16	$[-Y, 0, X]$	$\left[\frac{7}{8}, x+\frac{1}{4}, \frac{3}{8}\right]$	[30,-31]
17	$[-X, -Y, 0]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{1}{2}-x\right]$	[33,-36]
18	$[X, -Y, 0]$	$\left[\frac{3}{8}, \frac{3}{8}, x+\frac{3}{4}\right]$	[34,-35]
19	$[-X, 0, Y]$	$\left[\frac{5}{8}, \frac{3}{4}-x, \frac{1}{8}\right]$	[37,-40]
20	$[X, 0, Y]$	$\left[\frac{1}{8}, x, \frac{1}{8}\right]$	[38,-39]
21	$[0, -Y, X]$	$\left[\frac{3}{4}-x, \frac{1}{8}, \frac{5}{8}\right]$	[41,-44]
22	$[0, -Y, -X]$	$\left[x, \frac{1}{8}, \frac{1}{8}\right]$	[42,-43]
23	$[-Y, -X, 0]$	$\left[\frac{1}{8}, \frac{1}{8}, x\right]$	[45,-48]
24	$[-Y, X, 0]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{3}{4}-x\right]$	[46,-47]

Table 29: Wyckoff bond: 192e@48f, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{8}, \frac{1}{8}\right]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{3}{4}-x, \frac{1}{8}, \frac{5}{8}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{3}{4}-x, \frac{1}{8}, \frac{5}{8}\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{8}, \frac{1}{8}\right]$	[4]
5	$[Z, X, Y]$	$\left[\frac{1}{8}, x, \frac{1}{8}\right]$	[5]
6	$[Z, -X, -Y]$	$\left[\frac{5}{8}, \frac{3}{4}-x, \frac{1}{8}\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{5}{8}, \frac{3}{4}-x, \frac{1}{8}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{8}, x, \frac{1}{8}\right]$	[8]
9	$[Y, Z, X]$	$\left[\frac{1}{8}, \frac{1}{8}, x\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{3}{4}-x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{3}{4}-x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{8}, \frac{1}{8}, x\right]$	[12]
13	$[Y, X, -Z]$	$\left[\frac{7}{8}, x+\frac{1}{4}, \frac{3}{8}\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{7}{8}, -x, \frac{7}{8}\right]$	[14]
15	$[Y, -X, Z]$	$\left[\frac{7}{8}, -x, \frac{7}{8}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{7}{8}, x+\frac{1}{4}, \frac{3}{8}\right]$	[16]
17	$[X, Z, -Y]$	$\left[x+\frac{3}{4}, \frac{3}{8}, \frac{3}{8}\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{2}-x, \frac{7}{8}, \frac{3}{8}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}-x, \frac{7}{8}, \frac{3}{8}\right]$	[19]
20	$[X, -Z, Y]$	$\left[x+\frac{3}{4}, \frac{3}{8}, \frac{3}{8}\right]$	[20]

continued ...

Table 29

No.	vector	center	mapping
21	$[Z, Y, -X]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{1}{2} - x\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{3}{8}, \frac{3}{8}, x + \frac{3}{4}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{3}{8}, \frac{3}{8}, x + \frac{3}{4}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{1}{2} - x\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[\frac{1}{2} - x, \frac{7}{8}, \frac{3}{8}\right]$	[25]
26	$[X, Y, -Z]$	$\left[x + \frac{3}{4}, \frac{3}{8}, \frac{3}{8}\right]$	[26]
27	$[X, -Y, Z]$	$\left[x + \frac{3}{4}, \frac{3}{8}, \frac{3}{8}\right]$	[27]
28	$[-X, Y, Z]$	$\left[\frac{1}{2} - x, \frac{7}{8}, \frac{3}{8}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[\frac{7}{8}, -x, \frac{7}{8}\right]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{7}{8}, x + \frac{1}{4}, \frac{3}{8}\right]$	[30]
31	$[Z, X, -Y]$	$\left[\frac{7}{8}, x + \frac{1}{4}, \frac{3}{8}\right]$	[31]
32	$[Z, -X, Y]$	$\left[\frac{7}{8}, -x, \frac{7}{8}\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{1}{2} - x\right]$	[33]
34	$[Y, -Z, X]$	$\left[\frac{3}{8}, \frac{3}{8}, x + \frac{3}{4}\right]$	[34]
35	$[-Y, Z, X]$	$\left[\frac{3}{8}, \frac{3}{8}, x + \frac{3}{4}\right]$	[35]
36	$[Y, Z, -X]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{1}{2} - x\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{5}{8}, \frac{3}{4} - x, \frac{1}{8}\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{1}{8}, x, \frac{1}{8}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{8}, x, \frac{1}{8}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{5}{8}, \frac{3}{4} - x, \frac{1}{8}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{3}{4} - x, \frac{1}{8}, \frac{5}{8}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x, \frac{1}{8}, \frac{1}{8}\right]$	[42]
43	$[X, Z, Y]$	$\left[x, \frac{1}{8}, \frac{1}{8}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{3}{4} - x, \frac{1}{8}, \frac{5}{8}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{1}{8}, \frac{1}{8}, x\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{3}{4} - x\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{1}{8}, \frac{5}{8}, \frac{3}{4} - x\right]$	[47]
48	$[Z, Y, X]$	$\left[\frac{1}{8}, \frac{1}{8}, x\right]$	[48]

* Wyckoff site: 96g, site symmetry: ..m

Table 30: Wyckoff bond: 96a@96g, bond symmetry: ..m

No.	vector	center	mapping
1	$[X, X, Y]$	$[x, x, z]$	[1,38]
2	$[-X, -X, Y]$	$\left[\frac{3}{4} - x, \frac{1}{4} - x, z + \frac{1}{2}\right]$	[2,37]
3	$[-X, X, -Y]$	$\left[\frac{1}{4} - x, x + \frac{1}{2}, \frac{3}{4} - z\right]$	[3,39]
4	$[X, -X, -Y]$	$\left[x + \frac{1}{2}, \frac{3}{4} - x, \frac{1}{4} - z\right]$	[4,40]
5	$[Y, X, X]$	$[z, x, x]$	[5,48]
6	$[Y, -X, -X]$	$\left[z + \frac{1}{2}, \frac{3}{4} - x, \frac{1}{4} - x\right]$	[6,47]
7	$[-Y, -X, X]$	$\left[\frac{3}{4} - z, \frac{1}{4} - x, x + \frac{1}{2}\right]$	[7,45]
8	$[-Y, X, -X]$	$\left[\frac{1}{4} - z, x + \frac{1}{2}, \frac{3}{4} - x\right]$	[8,46]
9	$[X, Y, X]$	$[x, z, x]$	[9,43]
10	$[-X, Y, -X]$	$\left[\frac{1}{4} - x, z + \frac{1}{2}, \frac{3}{4} - x\right]$	[10,44]
11	$[X, -Y, -X]$	$\left[x + \frac{1}{2}, \frac{3}{4} - z, \frac{1}{4} - x\right]$	[11,42]
12	$[-X, -Y, X]$	$\left[\frac{3}{4} - x, \frac{1}{4} - z, x + \frac{1}{2}\right]$	[12,41]
13	$[X, X, -Y]$	$\left[x + \frac{3}{4}, x + \frac{1}{4}, \frac{1}{2} - z\right]$	[13,26]

continued ...

Table 30

No.	vector	center	mapping
14	$[-X, -X, -Y]$	$[-x, -x, -z]$	[14,25]
15	$[X, -X, Y]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, z+\frac{3}{4}\right]$	[15,27]
16	$[-X, X, Y]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, z+\frac{1}{4}\right]$	[16,28]
17	$[X, Y, -X]$	$\left[x+\frac{3}{4}, z+\frac{1}{4}, \frac{1}{2}-x\right]$	[17,36]
18	$[-X, Y, X]$	$\left[\frac{1}{2}-x, z+\frac{3}{4}, x+\frac{1}{4}\right]$	[18,35]
19	$[-X, -Y, -X]$	$[-x, -z, -x]$	[19,33]
20	$[X, -Y, X]$	$\left[x+\frac{1}{4}, \frac{1}{2}-z, x+\frac{3}{4}\right]$	[20,34]
21	$[Y, X, -X]$	$\left[z+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	[21,31]
22	$[Y, -X, X]$	$\left[z+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	[22,32]
23	$[-Y, X, X]$	$\left[\frac{1}{2}-z, x+\frac{3}{4}, x+\frac{1}{4}\right]$	[23,30]
24	$[-Y, -X, -X]$	$[-z, -x, -x]$	[24,29]

Table 31: Wyckoff bond: 96b@96g, bond symmetry: ..m

No.	vector	center	mapping
1	$[X, -X, 0]$	$[x, x, z]$	[1,-38]
2	$[-X, X, 0]$	$\left[\frac{3}{4}-x, \frac{1}{4}-x, z+\frac{1}{2}\right]$	[2,-37]
3	$[-X, -X, 0]$	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-z\right]$	[3,-39]
4	$[X, X, 0]$	$\left[x+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-z\right]$	[4,-40]
5	$[0, X, -X]$	$[z, x, x]$	[5,-48]
6	$[0, -X, X]$	$\left[z+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-x\right]$	[6,-47]
7	$[0, -X, -X]$	$\left[\frac{3}{4}-z, \frac{1}{4}-x, x+\frac{1}{2}\right]$	[7,-45]
8	$[0, X, X]$	$\left[\frac{1}{4}-z, x+\frac{1}{2}, \frac{3}{4}-x\right]$	[8,-46]
9	$[-X, 0, X]$	$[x, z, x]$	[9,-43]
10	$[X, 0, -X]$	$\left[\frac{1}{4}-x, z+\frac{1}{2}, \frac{3}{4}-x\right]$	[10,-44]
11	$[-X, 0, -X]$	$\left[x+\frac{1}{2}, \frac{3}{4}-z, \frac{1}{4}-x\right]$	[11,-42]
12	$[X, 0, X]$	$\left[\frac{3}{4}-x, \frac{1}{4}-z, x+\frac{1}{2}\right]$	[12,-41]
13	$[-X, X, 0]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-z\right]$	[13,-26]
14	$[X, -X, 0]$	$[-x, -x, -z]$	[14,-25]
15	$[-X, -X, 0]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, z+\frac{3}{4}\right]$	[15,-27]
16	$[X, X, 0]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, z+\frac{1}{4}\right]$	[16,-28]
17	$[X, 0, X]$	$\left[x+\frac{3}{4}, z+\frac{1}{4}, \frac{1}{2}-x\right]$	[17,-36]
18	$[-X, 0, -X]$	$\left[\frac{1}{2}-x, z+\frac{3}{4}, x+\frac{1}{4}\right]$	[18,-35]
19	$[-X, 0, X]$	$[-x, -z, -x]$	[19,-33]
20	$[X, 0, -X]$	$\left[x+\frac{1}{4}, \frac{1}{2}-z, x+\frac{3}{4}\right]$	[20,-34]
21	$[0, -X, -X]$	$\left[z+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	[21,-31]
22	$[0, X, X]$	$\left[z+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	[22,-32]
23	$[0, -X, X]$	$\left[\frac{1}{2}-z, x+\frac{3}{4}, x+\frac{1}{4}\right]$	[23,-30]
24	$[0, X, -X]$	$[-z, -x, -x]$	[24,-29]

Table 32: Wyckoff bond: 192c@96g, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, z]$	[1]

continued ...

Table 32

No.	vector	center	mapping
2	$[-X, -Y, Z]$	$\left[\frac{3}{4}-x, \frac{1}{4}-x, z+\frac{1}{2}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-z\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-z\right]$	[4]
5	$[Z, X, Y]$	$[z, x, x]$	[5]
6	$[Z, -X, -Y]$	$\left[z+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-x\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{3}{4}-z, \frac{1}{4}-x, x+\frac{1}{2}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{4}-z, x+\frac{1}{2}, \frac{3}{4}-x\right]$	[8]
9	$[Y, Z, X]$	$[x, z, x]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{4}-x, z+\frac{1}{2}, \frac{3}{4}-x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[x+\frac{1}{2}, \frac{3}{4}-z, \frac{1}{4}-x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{3}{4}-x, \frac{1}{4}-z, x+\frac{1}{2}\right]$	[12]
13	$[Y, X, -Z]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-z\right]$	[13]
14	$[-Y, -X, -Z]$	$[-x, -x, -z]$	[14]
15	$[Y, -X, Z]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, z+\frac{3}{4}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, z+\frac{1}{4}\right]$	[16]
17	$[X, Z, -Y]$	$\left[x+\frac{3}{4}, z+\frac{1}{4}, \frac{1}{2}-x\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{2}-x, z+\frac{3}{4}, x+\frac{1}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$[-x, -z, -x]$	[19]
20	$[X, -Z, Y]$	$\left[x+\frac{1}{4}, \frac{1}{2}-z, x+\frac{3}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[z+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	[21]
22	$[Z, -Y, X]$	$\left[z+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{2}-z, x+\frac{3}{4}, x+\frac{1}{4}\right]$	[23]
24	$[-Z, -Y, -X]$	$[-z, -x, -x]$	[24]
25	$[-X, -Y, -Z]$	$[-x, -x, -z]$	[25]
26	$[X, Y, -Z]$	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-z\right]$	[26]
27	$[X, -Y, Z]$	$\left[x+\frac{1}{4}, \frac{1}{2}-x, z+\frac{3}{4}\right]$	[27]
28	$[-X, Y, Z]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, z+\frac{1}{4}\right]$	[28]
29	$[-Z, -X, -Y]$	$[-z, -x, -x]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{1}{2}-z, x+\frac{3}{4}, x+\frac{1}{4}\right]$	[30]
31	$[Z, X, -Y]$	$\left[z+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	[31]
32	$[Z, -X, Y]$	$\left[z+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	[32]
33	$[-Y, -Z, -X]$	$[-x, -z, -x]$	[33]
34	$[Y, -Z, X]$	$\left[x+\frac{1}{4}, \frac{1}{2}-z, x+\frac{3}{4}\right]$	[34]
35	$[-Y, Z, X]$	$\left[\frac{1}{2}-x, z+\frac{3}{4}, x+\frac{1}{4}\right]$	[35]
36	$[Y, Z, -X]$	$\left[x+\frac{3}{4}, z+\frac{1}{4}, \frac{1}{2}-x\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{3}{4}-x, \frac{1}{4}-x, z+\frac{1}{2}\right]$	[37]
38	$[Y, X, Z]$	$[x, x, z]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-z\right]$	[39]
40	$[Y, -X, -Z]$	$\left[x+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-z\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{3}{4}-x, \frac{1}{4}-z, x+\frac{1}{2}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x+\frac{1}{2}, \frac{3}{4}-z, \frac{1}{4}-x\right]$	[42]
43	$[X, Z, Y]$	$[x, z, x]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{4}-x, z+\frac{1}{2}, \frac{3}{4}-x\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{3}{4}-z, \frac{1}{4}-x, x+\frac{1}{2}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{1}{4}-z, x+\frac{1}{2}, \frac{3}{4}-x\right]$	[46]
47	$[Z, -Y, -X]$	$\left[z+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-x\right]$	[47]
48	$[Z, Y, X]$	$[z, x, x]$	[48]

* Wyckoff site: `96h`, site symmetry: `..2`

Table 33: Wyckoff bond: `96a@96h`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y, X, X]$	$[0, y, -y]$	`[1,-19]`
2	$[-Y, -X, X]$	$[\frac{3}{4}, \frac{1}{4} - y, \frac{1}{2} - y]$	`[2,-17]`
3	$[-Y, X, -X]$	$[\frac{1}{4}, y + \frac{1}{2}, y + \frac{3}{4}]$	`[3,-20]`
4	$[Y, -X, -X]$	$[\frac{1}{2}, \frac{3}{4} - y, y + \frac{1}{4}]$	`[4,-18]`
5	$[X, Y, X]$	$[-y, 0, y]$	`[5,-14]`
6	$[X, -Y, -X]$	$[\frac{1}{2} - y, \frac{3}{4}, \frac{1}{4} - y]$	`[6,-16]`
7	$[-X, -Y, X]$	$[y + \frac{3}{4}, \frac{1}{4}, y + \frac{1}{2}]$	`[7,-13]`
8	$[-X, Y, -X]$	$[y + \frac{1}{4}, \frac{1}{2}, \frac{3}{4} - y]$	`[8,-15]`
9	$[X, X, Y]$	$[y, -y, 0]$	`[9,-24]`
10	$[-X, X, -Y]$	$[\frac{1}{4} - y, \frac{1}{2} - y, \frac{3}{4}]$	`[10,-22]`
11	$[X, -X, -Y]$	$[y + \frac{1}{2}, y + \frac{3}{4}, \frac{1}{4}]$	`[11,-23]`
12	$[-X, -X, Y]$	$[\frac{3}{4} - y, y + \frac{1}{4}, \frac{1}{2}]$	`[12,-21]`
13	$[-Y, -X, -X]$	$[0, -y, y]$	`[25,-43]`
14	$[Y, X, -X]$	$[\frac{1}{4}, y + \frac{3}{4}, y + \frac{1}{2}]$	`[26,-41]`
15	$[Y, -X, X]$	$[\frac{3}{4}, \frac{1}{2} - y, \frac{1}{4} - y]$	`[27,-44]`
16	$[-Y, X, X]$	$[\frac{1}{2}, y + \frac{1}{4}, \frac{3}{4} - y]$	`[28,-42]`
17	$[-X, -Y, -X]$	$[y, 0, -y]$	`[29,-38]`
18	$[-X, Y, X]$	$[y + \frac{1}{2}, \frac{1}{4}, y + \frac{3}{4}]$	`[30,-40]`
19	$[X, Y, -X]$	$[\frac{1}{4} - y, \frac{3}{4}, \frac{1}{2} - y]$	`[31,-37]`
20	$[X, -Y, X]$	$[\frac{3}{4} - y, \frac{1}{2}, y + \frac{1}{4}]$	`[32,-39]`
21	$[-X, -X, -Y]$	$[-y, y, 0]$	`[33,-48]`
22	$[X, -X, Y]$	$[y + \frac{3}{4}, y + \frac{1}{2}, \frac{1}{4}]$	`[34,-46]`
23	$[-X, X, Y]$	$[\frac{1}{2} - y, \frac{1}{4} - y, \frac{3}{4}]$	`[35,-47]`
24	$[X, X, -Y]$	$[y + \frac{1}{4}, \frac{3}{4} - y, \frac{1}{2}]$	`[36,-45]`

Table 34: Wyckoff bond: `96b@96h`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[0, X, -X]$	$[0, y, -y]$	`[1,19]`
2	$[0, -X, -X]$	$[\frac{3}{4}, \frac{1}{4} - y, \frac{1}{2} - y]$	`[2,17]`
3	$[0, X, X]$	$[\frac{1}{4}, y + \frac{1}{2}, y + \frac{3}{4}]$	`[3,20]`
4	$[0, -X, X]$	$[\frac{1}{2}, \frac{3}{4} - y, y + \frac{1}{4}]$	`[4,18]`
5	$[-X, 0, X]$	$[-y, 0, y]$	`[5,14]`
6	$[-X, 0, -X]$	$[\frac{1}{2} - y, \frac{3}{4}, \frac{1}{4} - y]$	`[6,16]`
7	$[X, 0, X]$	$[y + \frac{3}{4}, \frac{1}{4}, y + \frac{1}{2}]$	`[7,13]`
8	$[X, 0, -X]$	$[y + \frac{1}{4}, \frac{1}{2}, \frac{3}{4} - y]$	`[8,15]`
9	$[X, -X, 0]$	$[y, -y, 0]$	`[9,24]`
10	$[-X, -X, 0]$	$[\frac{1}{4} - y, \frac{1}{2} - y, \frac{3}{4}]$	`[10,22]`
11	$[X, X, 0]$	$[y + \frac{1}{2}, y + \frac{3}{4}, \frac{1}{4}]$	`[11,23]`
12	$[-X, X, 0]$	$[\frac{3}{4} - y, y + \frac{1}{4}, \frac{1}{2}]$	`[12,21]`
13	$[0, -X, X]$	$[0, -y, y]$	`[25,43]`
14	$[0, X, X]$	$[\frac{1}{4}, y + \frac{3}{4}, y + \frac{1}{2}]$	`[26,41]`
15	$[0, -X, -X]$	$[\frac{3}{4}, \frac{1}{2} - y, \frac{1}{4} - y]$	`[27,44]`
16	$[0, X, -X]$	$[\frac{1}{2}, y + \frac{1}{4}, \frac{3}{4} - y]$	`[28,42]`

continued ...

Table 34

No.	vector	center	mapping
17	$[X, 0, -X]$	$[y, 0, -y]$	[29,38]
18	$[X, 0, X]$	$[y+\frac{1}{2}, \frac{1}{4}, y+\frac{3}{4}]$	[30,40]
19	$[-X, 0, -X]$	$[\frac{1}{4}-y, \frac{3}{4}, \frac{1}{2}-y]$	[31,37]
20	$[-X, 0, X]$	$[\frac{3}{4}-y, \frac{1}{2}, y+\frac{1}{4}]$	[32,39]
21	$[-X, X, 0]$	$[-y, y, 0]$	[33,48]
22	$[X, X, 0]$	$[y+\frac{3}{4}, y+\frac{1}{2}, \frac{1}{4}]$	[34,46]
23	$[-X, -X, 0]$	$[\frac{1}{2}-y, \frac{1}{4}-y, \frac{3}{4}]$	[35,47]
24	$[X, -X, 0]$	$[y+\frac{1}{4}, \frac{3}{4}-y, \frac{1}{2}]$	[36,45]

Table 35: Wyckoff bond: 192c@96h, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, y, -y]$	[1]
2	$[-X, -Y, Z]$	$[\frac{3}{4}, \frac{1}{4}-y, \frac{1}{2}-y]$	[2]
3	$[-X, Y, -Z]$	$[\frac{1}{4}, y+\frac{1}{2}, y+\frac{3}{4}]$	[3]
4	$[X, -Y, -Z]$	$[\frac{1}{2}, \frac{3}{4}-y, y+\frac{1}{4}]$	[4]
5	$[Z, X, Y]$	$[-y, 0, y]$	[5]
6	$[Z, -X, -Y]$	$[\frac{1}{2}-y, \frac{3}{4}, \frac{1}{4}-y]$	[6]
7	$[-Z, -X, Y]$	$[y+\frac{3}{4}, \frac{1}{4}, y+\frac{1}{2}]$	[7]
8	$[-Z, X, -Y]$	$[y+\frac{1}{4}, \frac{1}{2}, \frac{3}{4}-y]$	[8]
9	$[Y, Z, X]$	$[y, -y, 0]$	[9]
10	$[-Y, Z, -X]$	$[\frac{1}{4}-y, \frac{1}{2}-y, \frac{3}{4}]$	[10]
11	$[Y, -Z, -X]$	$[y+\frac{1}{2}, y+\frac{3}{4}, \frac{1}{4}]$	[11]
12	$[-Y, -Z, X]$	$[\frac{3}{4}-y, y+\frac{1}{4}, \frac{1}{2}]$	[12]
13	$[Y, X, -Z]$	$[y+\frac{3}{4}, \frac{1}{4}, y+\frac{1}{2}]$	[13]
14	$[-Y, -X, -Z]$	$[-y, 0, y]$	[14]
15	$[Y, -X, Z]$	$[y+\frac{1}{4}, \frac{1}{2}, \frac{3}{4}-y]$	[15]
16	$[-Y, X, Z]$	$[\frac{1}{2}-y, \frac{3}{4}, \frac{1}{4}-y]$	[16]
17	$[X, Z, -Y]$	$[\frac{3}{4}, \frac{1}{4}-y, \frac{1}{2}-y]$	[17]
18	$[-X, Z, Y]$	$[\frac{1}{2}, \frac{3}{4}-y, y+\frac{1}{4}]$	[18]
19	$[-X, -Z, -Y]$	$[0, y, -y]$	[19]
20	$[X, -Z, Y]$	$[\frac{1}{4}, y+\frac{1}{2}, y+\frac{3}{4}]$	[20]
21	$[Z, Y, -X]$	$[\frac{3}{4}-y, y+\frac{1}{4}, \frac{1}{2}]$	[21]
22	$[Z, -Y, X]$	$[\frac{1}{4}-y, \frac{1}{2}-y, \frac{3}{4}]$	[22]
23	$[-Z, Y, X]$	$[y+\frac{1}{2}, y+\frac{3}{4}, \frac{1}{4}]$	[23]
24	$[-Z, -Y, -X]$	$[y, -y, 0]$	[24]
25	$[-X, -Y, -Z]$	$[0, -y, y]$	[25]
26	$[X, Y, -Z]$	$[\frac{1}{4}, y+\frac{3}{4}, y+\frac{1}{2}]$	[26]
27	$[X, -Y, Z]$	$[\frac{3}{4}, \frac{1}{2}-y, \frac{1}{4}-y]$	[27]
28	$[-X, Y, Z]$	$[\frac{1}{2}, y+\frac{1}{4}, \frac{3}{4}-y]$	[28]
29	$[-Z, -X, -Y]$	$[y, 0, -y]$	[29]
30	$[-Z, X, Y]$	$[y+\frac{1}{2}, \frac{1}{4}, y+\frac{3}{4}]$	[30]
31	$[Z, X, -Y]$	$[\frac{1}{4}-y, \frac{3}{4}, \frac{1}{2}-y]$	[31]
32	$[Z, -X, Y]$	$[\frac{3}{4}-y, \frac{1}{2}, y+\frac{1}{4}]$	[32]
33	$[-Y, -Z, -X]$	$[-y, y, 0]$	[33]

continued ...

Table 35

No.	vector	center	mapping
34	$[Y, -Z, X]$	$\left[y+\frac{3}{4}, y+\frac{1}{2}, \frac{1}{4}\right]$	[34]
35	$[-Y, Z, X]$	$\left[\frac{1}{2}-y, \frac{1}{4}-y, \frac{3}{4}\right]$	[35]
36	$[Y, Z, -X]$	$\left[y+\frac{1}{4}, \frac{3}{4}-y, \frac{1}{2}\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{4}-y, \frac{3}{4}, \frac{1}{2}-y\right]$	[37]
38	$[Y, X, Z]$	$[y, 0, -y]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{3}{4}-y, \frac{1}{2}, y+\frac{1}{4}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[y+\frac{1}{2}, \frac{1}{4}, y+\frac{3}{4}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{4}, y+\frac{3}{4}, y+\frac{1}{2}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[\frac{1}{2}, y+\frac{1}{4}, \frac{3}{4}-y\right]$	[42]
43	$[X, Z, Y]$	$[0, -y, y]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{3}{4}, \frac{1}{2}-y, \frac{1}{4}-y\right]$	[44]
45	$[-Z, -Y, X]$	$\left[y+\frac{1}{4}, \frac{3}{4}-y, \frac{1}{2}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[y+\frac{3}{4}, y+\frac{1}{2}, \frac{1}{4}\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{1}{2}-y, \frac{1}{4}-y, \frac{3}{4}\right]$	[47]
48	$[Z, Y, X]$	$[-y, y, 0]$	[48]

* Wyckoff site: 192i, site symmetry: 1

Table 36: Wyckoff bond: 192a@192i, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{3}{4}-x, \frac{1}{4}-y, z+\frac{1}{2}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{1}{4}-x, y+\frac{1}{2}, \frac{3}{4}-z\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x+\frac{1}{2}, \frac{3}{4}-y, \frac{1}{4}-z\right]$	[4]
5	$[Z, X, Y]$	$[z, x, y]$	[5]
6	$[Z, -X, -Y]$	$\left[z+\frac{1}{2}, \frac{3}{4}-x, \frac{1}{4}-y\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{3}{4}-z, \frac{1}{4}-x, y+\frac{1}{2}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{4}-z, x+\frac{1}{2}, \frac{3}{4}-y\right]$	[8]
9	$[Y, Z, X]$	$[y, z, x]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{4}-y, z+\frac{1}{2}, \frac{3}{4}-x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[y+\frac{1}{2}, \frac{3}{4}-z, \frac{1}{4}-x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{3}{4}-y, \frac{1}{4}-z, x+\frac{1}{2}\right]$	[12]
13	$[Y, X, -Z]$	$\left[y+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-z\right]$	[13]
14	$[-Y, -X, -Z]$	$[-y, -x, -z]$	[14]
15	$[Y, -X, Z]$	$\left[y+\frac{1}{4}, \frac{1}{2}-x, z+\frac{3}{4}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{2}-y, x+\frac{3}{4}, z+\frac{1}{4}\right]$	[16]
17	$[X, Z, -Y]$	$\left[x+\frac{3}{4}, z+\frac{1}{4}, \frac{1}{2}-y\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{2}-x, z+\frac{3}{4}, y+\frac{1}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$[-x, -z, -y]$	[19]
20	$[X, -Z, Y]$	$\left[x+\frac{1}{4}, \frac{1}{2}-z, y+\frac{3}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[z+\frac{3}{4}, y+\frac{1}{4}, \frac{1}{2}-x\right]$	[21]
22	$[Z, -Y, X]$	$\left[z+\frac{1}{4}, \frac{1}{2}-y, x+\frac{3}{4}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{2}-z, y+\frac{3}{4}, x+\frac{1}{4}\right]$	[23]
24	$[-Z, -Y, -X]$	$[-z, -y, -x]$	[24]
25	$[-X, -Y, -Z]$	$[-x, -y, -z]$	[25]
26	$[X, Y, -Z]$	$\left[x+\frac{1}{4}, y+\frac{3}{4}, \frac{1}{2}-z\right]$	[26]

continued ...

Table 36

No.	vector	center	mapping
27	$[X, -Y, Z]$	$\left[x+\frac{3}{4}, \frac{1}{2}-y, z+\frac{1}{4}\right]$	[27]
28	$[-X, Y, Z]$	$\left[\frac{1}{2}-x, y+\frac{1}{4}, z+\frac{3}{4}\right]$	[28]
29	$[-Z, -X, -Y]$	$[-z, -x, -y]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{1}{2}-z, x+\frac{1}{4}, y+\frac{3}{4}\right]$	[30]
31	$[Z, X, -Y]$	$\left[z+\frac{1}{4}, x+\frac{3}{4}, \frac{1}{2}-y\right]$	[31]
32	$[Z, -X, Y]$	$\left[z+\frac{3}{4}, \frac{1}{2}-x, y+\frac{1}{4}\right]$	[32]
33	$[-Y, -Z, -X]$	$[-y, -z, -x]$	[33]
34	$[Y, -Z, X]$	$\left[y+\frac{3}{4}, \frac{1}{2}-z, x+\frac{1}{4}\right]$	[34]
35	$[-Y, Z, X]$	$\left[\frac{1}{2}-y, z+\frac{1}{4}, x+\frac{3}{4}\right]$	[35]
36	$[Y, Z, -X]$	$\left[y+\frac{1}{4}, z+\frac{3}{4}, \frac{1}{2}-x\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{4}-y, \frac{3}{4}-x, z+\frac{1}{2}\right]$	[37]
38	$[Y, X, Z]$	$[y, x, z]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{3}{4}-y, x+\frac{1}{2}, \frac{1}{4}-z\right]$	[39]
40	$[Y, -X, -Z]$	$\left[y+\frac{1}{2}, \frac{1}{4}-x, \frac{3}{4}-z\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{4}-x, \frac{3}{4}-z, y+\frac{1}{2}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x+\frac{1}{2}, \frac{1}{4}-z, \frac{3}{4}-y\right]$	[42]
43	$[X, Z, Y]$	$[x, z, y]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{3}{4}-x, z+\frac{1}{2}, \frac{1}{4}-y\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{1}{4}-z, \frac{3}{4}-y, x+\frac{1}{2}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{3}{4}-z, y+\frac{1}{2}, \frac{1}{4}-x\right]$	[46]
47	$[Z, -Y, -X]$	$\left[z+\frac{1}{2}, \frac{1}{4}-y, \frac{3}{4}-x\right]$	[47]
48	$[Z, Y, X]$	$[z, y, x]$	[48]